

Three-Dimensional Small-Disturbance Solutions

In this chapter, three-dimensional small-disturbance solutions will be derived for some simple cases such as the large aspect ratio wing, the slender pointed wing, and the slender cylindrical body. Flow problems requiring more detailed geometries will be treated in the forthcoming chapters.

8.1 Finite Wing: The Lifting Line Model

The three-dimensional lifting wing problem was formulated in Chapter 4, and it is clear that obtaining an analytic solution of the integral equations is difficult. However, it is possible to approximate the lifting properties of a wing by a single lifting line, an approximation that will allow a closed-form solution. In spite of the considerable simplifications in this model it captures the basic features of three-dimensional lifting flows and predicts the reduction of lift slope and the increase in induced drag with decreasing aspect ratio.

8.1.1 Definition of the Problem

Consider a lifting, thin, finite wing (described in Section 4.5 and shown in Fig. 8.1) that is moving at a constant speed in an otherwise undisturbed fluid. The free stream of speed Q_∞ has a small angle of attack α , relative to the coordinate system attached to the wing.

The velocity field for this potential flow problem can be obtained by solving Laplace's equation for the perturbation potential Φ :

$$\nabla^2 \Phi = 0 \quad (8.1)$$

Following Section 4.5, the boundary condition requiring no flow across the wing solid surface will be approximated at $z = 0$, for the case of small angle of attack, by

$$\frac{\partial \Phi}{\partial z}(x, y, 0\pm) = Q_\infty \left(\frac{\partial \eta}{\partial x} - \alpha \right) \quad (8.2)$$

where $\eta = \eta_c(x, y)$ is the camber surface placed on the x - y plane and for simplicity the subscript c is omitted in this chapter. For modeling the lifting surface, a vortex distribution is selected (as formulated in Section 4.5). The unknown vortex distribution $\gamma_x(x, y)$ and $\gamma_y(x, y)$ (shown in Fig. 4.9) is placed on the wing's projected area at the $z = 0$ plane. The resulting integral equation is

$$\frac{-1}{4\pi} \int_{\text{wing+wake}} \frac{\gamma_y(x - x_0) - \gamma_x(y - y_0)}{[(x - x_0)^2 + (y - y_0)^2]^{3/2}} dx_0 dy_0 = Q_\infty \left(\frac{\partial \eta}{\partial x} - \alpha \right) \quad (8.3)$$

which should hold for any point on the wing. A proper (and unique) solution for the vortex distribution will have to fulfill the Kutta condition along the trailing edge, such that the vorticity component parallel to the trailing edge ($\gamma_{\text{T.E.}}$) is zero:

$$\gamma_{\text{T.E.}} = 0 \quad (8.4)$$

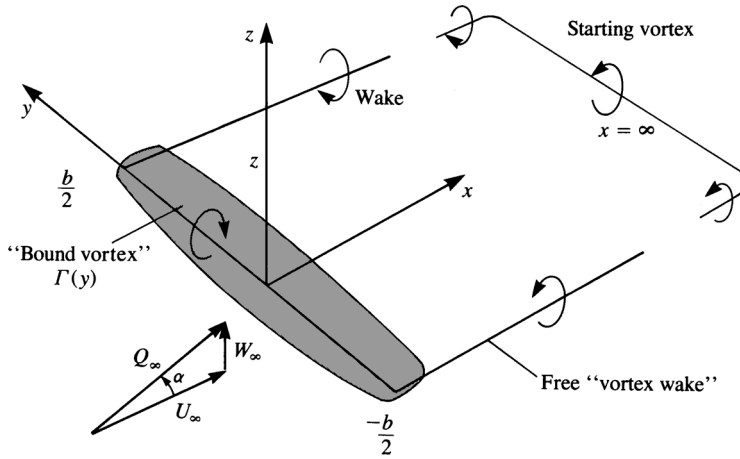


Figure 8.1 Far field horseshoe model of a finite wing.

Also, since vortex lines do not begin or end in a fluid (Eq. (4.64)), the solution must comply with

$$\left| \frac{\partial \gamma_x}{\partial x} \right| = \left| \frac{\partial \gamma_y}{\partial y} \right| \quad (8.5)$$

8.1.2 The Lifting-Line Model

The simplest model that can be suggested to solve this problem is where the chordwise circulation, at any spanwise station, is replaced by a single concentrated vortex. Also, these local vortices of circulation $\Gamma(y)$ will be placed along a single spanwise line. Based on the results of Section 5.5 for the two-dimensional lumped-vortex element, this vortex line will be placed at the wing's quarter-chord line along the span, $-b/2 < y < b/2$ (this bound vortex line is assumed to be straight and parallel to the y axis, as shown in Fig. 8.1). The above positioning of the vortex line at the wing's quarter-chord line effectively satisfies the Kutta condition of Eq. (8.4) as was shown in Section 5.5.

At this point, attention needs to be focused on the vortex theorems requiring that a vortex line cannot start or end abruptly in a fluid (or Eq. (8.5)). Therefore, if any change of the vortex line strength $d\Gamma(y)/dy$ is introduced, it must be followed by introducing a similar vorticity component in the other direction γ_x . In other words, the vortex line does not terminate at this point but it changes direction, and its strength remains constant.

The most physical application of these principles is to “shed” these trailing vortices into the flow and create a “wake” such that there will be no force acting on these free vortices. Following Section 4.7, this requirement reduces to the condition that the flow along this segment must be parallel to Γ (where positive Γ is according to the right-hand rule):

$$\mathbf{q} \times \Gamma_{\text{wake}} = 0 \quad (8.6)$$

The most basic element that will fit these requirements will have the shape of a “horseshoe” vortex (Fig. 8.1), which will have constant “bound vorticity” Γ along its quarter-chord line, will turn backward at the wing tips and will continue far behind the wing, and eventually will be closed by the starting vortex. It is assumed here that the flow is steady and therefore the starting vortex is far downstream and its influence can be neglected.

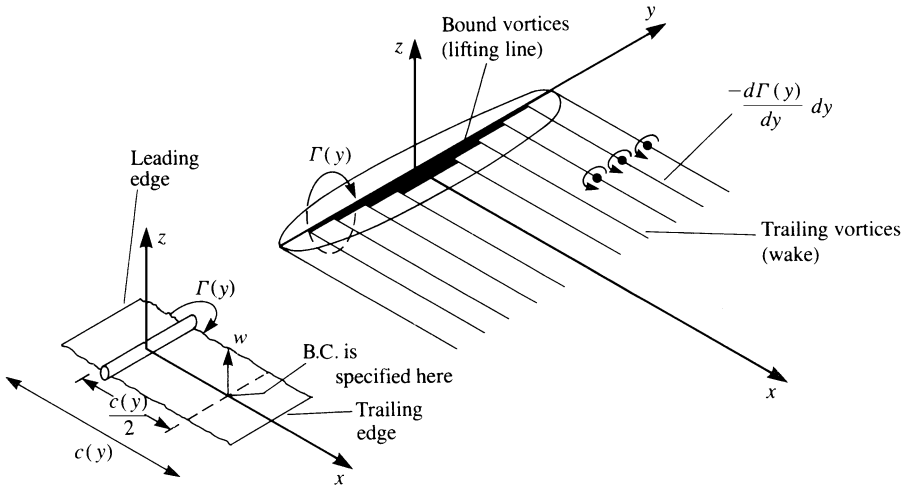


Figure 8.2 Lifting-line model consisting of horseshoe vortices. The bound vortex segment of all vortices is placed on the y axis.

A more refined model of the finite wing was first proposed by the German scientist Ludwig Prandtl (see Ref. 5.2) during World War 1 and it uses a large number of such spanwise horseshoe vortices, as illustrated by Fig. 8.2 (the following analysis is in the spirit of this early model). The straight bound vortex $\Gamma(y)$ in this case is placed along the y axis and at each spanwise station the leading edge is one-quarter chord ahead of this line and the local trailing edge is three-quarter chord behind the vortex line. Now, let us examine the integral equation (Eq. (8.3)) for the case of the flat lifting surface, where $\partial\eta/\partial x = 0$. The equation now simply states the boundary condition of Eq. (8.2):

$$\frac{\partial\Phi_{\text{wing}}}{\partial z} + \frac{\partial\Phi_{\text{wake}}}{\partial z} + Q_{\infty}\alpha = 0 \quad (8.2a)$$

That is, the sum of the normal velocity components induced by the wing $w_b = \partial\Phi_{\text{wing}}/\partial z$ and wake vortices $w_i = \partial\Phi_{\text{wake}}/\partial z$, plus the normal velocity component of the free-stream flow $Q_{\infty}\alpha$, must be zero on the wing's solid boundary:

$$w_b + w_i + Q_{\infty}\alpha = 0 \quad (8.7)$$

where w is considered to be positive in the $+z$ direction. The subscripts $()_b$ and $()_i$ stand for bound (on wing) and induced (by wake) influences, respectively.

The velocity component w_b induced by the lifting line on the section with a chord $c(y)$ can be estimated by using the lumped-vortex model with the downwash calculated at the collocation point located at the $3/4$ chord. Consider the spanwise component ($-y_0 \leq y \leq y_0$) of a typical horseshoe vortex in Fig. 8.3 with strength $\Delta\Gamma(y_0)$ (where $\Delta\Gamma(y_0) = -(d\Gamma(y_0)/dy)dy_0$). The downwash Δw_b at the collocation point $(c/2, y)$ due to this segment is given by Eq. (2.69) (see Fig. 2.15, which defines the angles β in this formula) as

$$\begin{aligned} \Delta w_b &= \frac{-\Delta\Gamma}{4\pi d}(\cos\beta_1 - \cos\beta_2) \\ &= \frac{-\Delta\Gamma}{4\pi[c(y)/2]} \left[\frac{y + y_0}{\sqrt{(c/2)^2 + (y + y_0)^2}} + \frac{y_0 - y}{\sqrt{(c/2)^2 + (y_0 - y)^2}} \right] \end{aligned}$$

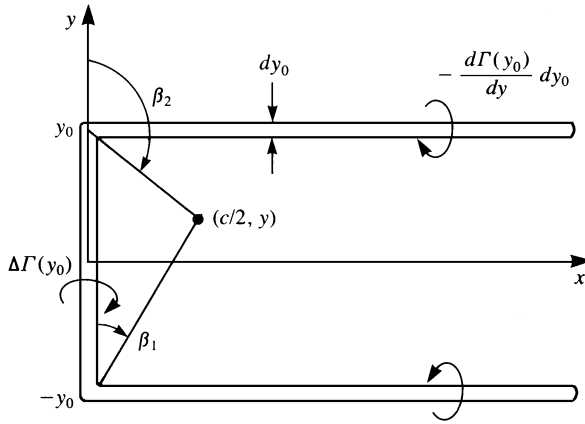


Figure 8.3 Velocity induced by the segments of a typical horseshoe element.

For a wing of large aspect ratio, we can neglect $(c/2)^2$ in the square root terms to get

$$\Delta w_b = \frac{-\Delta\Gamma(y_0)}{4\pi[c(y)/2]}[1 + 1]$$

The result for the complete lifting line (evaluated at y) is obtained by summing the results for all the horseshoe vortices and is

$$w_b = \frac{-\Gamma(y)}{2\pi[c(y)/2]} \quad (8.8)$$

Note that this is identical to the result obtained by applying a locally two-dimensional lumped-vortex model at each spanwise station, where the downwash w_b is measured at the 3/4 chord due to a vortex $\Gamma(y)$ placed at the 1/4 chord position (see inset in the left-hand side of Fig. 8.2).

Next, the downwash due to the wing trailing vortices must be evaluated. Since a change in the spanwise circulation $\Gamma(y)$ is allowed, and since no vortex can begin or end in the flow, the local change in this circulation is shed into the wake. Thus, the wake is now constructed from semi-infinite vortex lines with the strength of $(d\Gamma/dy)dy$ (Fig. 8.2). Before we proceed with the solution, the velocity induced by a single, semi-infinite trailing vortex line with a strength of $\Delta\Gamma = -(d\Gamma(y_0)/dy)dy_0$ is evaluated (note that for positive $\Delta\Gamma$ on the $+y$ side of the wing, negative $d\Gamma/dy$ is needed). The right-hand-side wake vortex line is located at a spanwise location y_0 , as shown in Fig. 8.3, and the downwash induced by this vortex at the collocation point $(c/2, y)$ is given by the result for a semi-infinite vortex line from Eq. (2.71). Since for a large aspect ratio wing $\beta_1 \approx \pi/2$, $\beta_2 \approx \pi$ (in Fig. 2.15) and therefore

$$w(y) = \frac{\Delta\Gamma(y_0)}{4\pi} \frac{1}{(y - y_0)} \quad (8.9)$$

which is exactly one half of the velocity induced by an infinite (two-dimensional) vortex of strength $\Delta\Gamma(y_0)$. With the aid of this equation the normal velocity component induced by

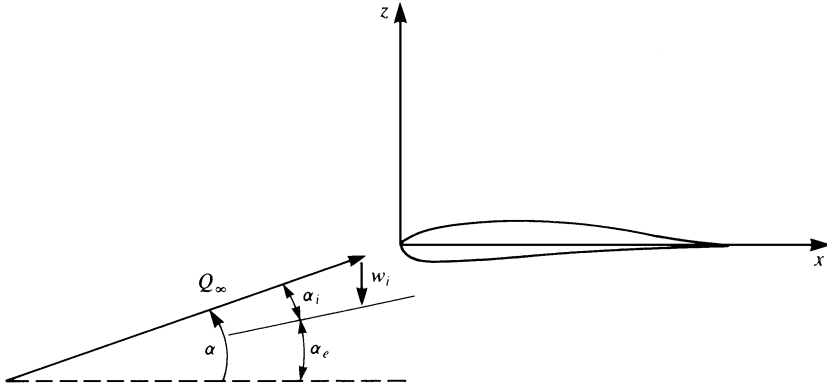


Figure 8.4 Two-dimensional section (in the $y = \text{const.}$ plane) of a three-dimensional wing. The angle of attack α is reduced by the induced downwash of the trailing vortices by α_i .

the trailing vortices of the wing becomes

$$w_i = \frac{1}{4\pi} \int_{-b/2}^{b/2} \frac{[-d\Gamma(y_0)/dy] dy_0}{y - y_0} \quad (8.10)$$

(Note that since the trailing vortices are assumed to lie in the $z = 0$ plane, their induced spanwise velocity component is zero from the Biot–Savart law, Eq. (2.68b).) Assuming that the wing aspect ratio is large ($b/c(y) \gg 1$) has allowed us to treat a spanwise station as a *two-dimensional* section and to transfer the boundary condition to the local three-quarter chord. Substituting Eqs. (8.8) and (8.10) into Eq. (8.7) yields

$$\frac{-\Gamma(y)}{2\pi[c(y)/2]} - \frac{1}{4\pi} \int_{-b/2}^{b/2} \frac{[d\Gamma(y_0)/dy] dy_0}{y - y_0} + Q_\infty \alpha = 0 \quad (8.11)$$

Dividing Eq. (8.11) by the free-stream speed Q_∞ results in

$$\frac{-\Gamma(y)}{\pi c(y) Q_\infty} - \frac{1}{4\pi Q_\infty} \int_{-b/2}^{b/2} \frac{[d\Gamma(y_0)/dy] dy_0}{y - y_0} + \alpha = 0 \quad (8.11a)$$

This is the Prandtl lifting-line integrodifferential equation for the spanwise load distribution $\Gamma(y)$. This equation can be viewed as a combination of the angles (as shown in Fig. 8.4):

$$-\alpha_e - \alpha_i + \alpha = 0 \quad (8.12)$$

where the induced downwash angle is (note that w is positive in the positive z direction)

$$\alpha_i \approx \frac{-w_i}{Q_\infty} \quad (8.13)$$

Equation (8.12) can be rearranged as

$$\alpha_e = \alpha - \alpha_i \quad (8.12a)$$

This means that in the case of the finite wing the effective angle of attack of a wing section α_e (the angle between the modified free-stream velocity $\mathbf{q}$ in Fig. 8.4 and the chord) is smaller than the actual geometric angle of attack α by α_i , which is a result of the downwash induced by the wake.

It is possible to generalize the result of this equation by assuming that the two-dimensional section has a local lift slope of m_0 and its local effective angle of attack is α_e . Now, if camber effects are to be accounted for too, then this angle is measured from the zero-lift angle of the section, such that

$$C_l(y) = \frac{\rho Q_\infty \Gamma(y)}{(1/2)\rho Q_\infty^2 c(y)} = m_0(y)\alpha_e(y) \quad (8.14)$$

Consequently, Eq. (8.12a) becomes

$$\alpha_e = \alpha - \alpha_i - \alpha_{L0} \quad (8.15)$$

where α_{L0} is the angle of zero lift due to the section camber (for cambered airfoils, usually α_{L0} is a negative number). A more general form of Eq. (8.11a) that allows for section camber and wing twist $\alpha(y)$ is now

$$\frac{-2\Gamma(y)}{m_0(y)c(y)Q_\infty} - \frac{1}{4\pi Q_\infty} \int_{-b/2}^{b/2} \frac{[d\Gamma(y_0)/dy] dy_0}{y - y_0} + \alpha(y) - \alpha_{L0}(y) = 0 \quad (8.16)$$

In this equation $\alpha(y)$ is the local angle of attack relative to Q_∞ and $\alpha_{L0}(y)$ is the airfoil section zero-lift angle. If it is assumed that these geometrical quantities are known, then $\Gamma(y)$ becomes the unknown in this equation. Also, at the wing tips the pressure difference [or the lift $\rho Q_\infty \Gamma(y = \pm b/2)$] must reduce to zero, that is,

$$\Gamma\left(y = \pm \frac{b}{2}\right) = 0 \quad (8.17)$$

8.1.3 The Aerodynamic Loads

The solution of Eq. (8.16) will provide the spanwise bound circulation distribution $\Gamma(y)$. To obtain the aerodynamic forces, the two-dimensional Kutta–Joukowski theorem will be applied (in the $y = \text{const.}$ plane). However, because of the wake-induced velocity, the free-stream vector will be rotated by $\alpha_i(y)$, as shown in Fig. 8.5. This angle can be calculated for a known $\Gamma(y)$ by using Eqs. (8.10) and (8.13):

$$\alpha_i = \frac{1}{4\pi Q_\infty} \int_{-b/2}^{b/2} \frac{[d\Gamma(y_0)/dy] dy_0}{y - y_0} \quad (8.18)$$

If we assume that α_i is small, then $\cos \alpha_i \approx 1$ and $\sin \alpha_i \approx \alpha_i$, and the lift of the wing is given by an integration of the local two-dimensional lift (see Eq. (3.113)) as

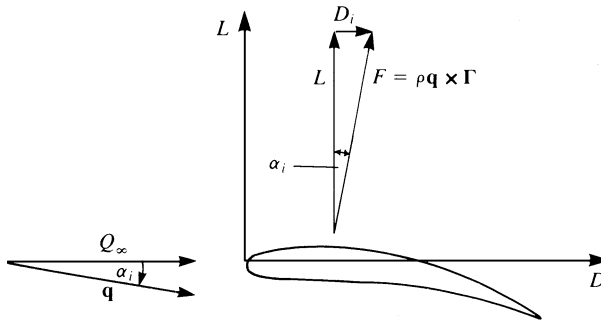


Figure 8.5 Tilting of the local (of section $y = \text{const.}$) lift vector by the angle α_i induced by the trailing vortices.

$$L = \rho Q_\infty \int_{-b/2}^{b/2} \Gamma(y) dy \quad (8.19)$$

while the drag force, which is created by turning the two-dimensional lift vector by the wake-induced flow, becomes

$$D_i = \rho Q_\infty \int_{-b/2}^{b/2} \alpha_i(y) \Gamma(y) dy \quad (8.20)$$

This drag is called *induced drag* because it is induced by the trailing vortices. This finite wing's drag is directly related to the lift and will diminish as the wingspan increases ($b \rightarrow \infty$). Equation (8.20) can also be rewritten in terms of the wake-induced downwash w_i :

$$D_i = -\rho \int_{-b/2}^{b/2} w_i(y) \Gamma(y) dy \quad (8.20a)$$

From the engineering point of view, the total drag D of a wing includes the induced drag D_i and the viscous drag D_0 :

$$D = D_i + D_0$$

For example, the two-dimensional viscous drag of a NACA 0009 section is presented in Fig. 5.20.

8.1.4 The Elliptic Lift Distribution

The spanwise circulation distribution $\Gamma(y)$ for a given planform shape can be obtained by solving Eq. (8.16). In the particular case of an elliptic distribution of the circulation, the solution becomes rather simple since the downwash w_i becomes constant along the wing span. Also, as will be shown later, wings having such a spanwise distribution will have minimum induced drag. The proposed distribution of $\Gamma(y)$ is shown in Fig. 8.6 and is

$$\Gamma(y) = \Gamma_{\max} \left[1 - \left(\frac{y}{b/2} \right)^2 \right]^{1/2} \quad (8.21)$$

This must be substituted into Eq. (8.16) so that the constant $\Gamma_{\max}$ can be evaluated. For simplicity, let us first calculate the downwash integral (second term in Eq. (8.16)). The term

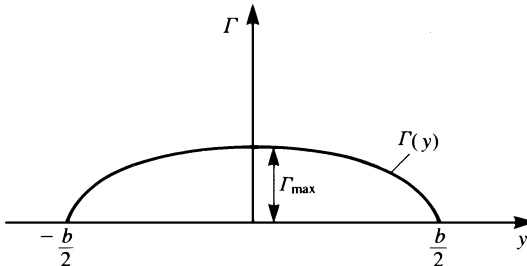


Figure 8.6 Elliptic spanwise distribution of the circulation $\Gamma(y)$.

$d\Gamma(y)/dy$ is evaluated by differentiating Eq. (8.21):

$$\frac{d\Gamma(y)}{dy} = \frac{\Gamma_{\max}}{2} \left[1 - \left(\frac{y}{b/2} \right)^2 \right]^{-1/2} \left(-2 \frac{4}{b^2} y \right)$$

and the downwash w_i is obtained by substituting this result into Eq. (8.10):

$$w_i(y) = \frac{\Gamma_{\max}}{\pi b^2} \int_{-b/2}^{b/2} \left[1 - \left(\frac{y_0}{b/2} \right)^2 \right]^{-1/2} \frac{y_0}{y - y_0} dy_0 \quad (8.22)$$

Note that when $y = y_0$, this integral is singular and therefore must be evaluated based on Cauchy's principal value. It is possible to arrive at Glauert's integral (Eq. (5.22)) by the transformation

$$y = \frac{b}{2} \cos \theta \quad (8.23)$$

$$dy = -\frac{b}{2} \sin \theta d\theta \quad (8.23a)$$

and at the wingtips $y = -b/2$, $\theta = \pi$ and at $y = b/2$, $\theta = 0$. This reduces Eq. (8.21) to

$$\Gamma(\theta) = \Gamma_{\max} [1 - \cos^2 \theta]^{1/2} = \Gamma_{\max} \sin \theta \quad (8.21a)$$

Substituting Eq. (8.23) into Eq. (8.22) yields

$$w_i = \frac{\Gamma_{\max}}{\pi b^2} \int_{\pi}^0 [1 - \cos^2 \theta_0]^{-1/2} \frac{(b/2) \cos \theta_0 [(-b/2) \sin \theta_0] d\theta_0}{(b/2)(\cos \theta - \cos \theta_0)}$$

The principal value of this integral can be obtained by using the Glauert integral, Eq. (5.22):

$$w_i = \frac{-\Gamma_{\max}}{2\pi b} \int_0^{\pi} \frac{\cos \theta_0 d\theta_0}{(\cos \theta_0 - \cos \theta)} = \frac{-\Gamma_{\max}}{2\pi b} \frac{\pi \sin \theta}{\sin \theta}$$

Consequently, w_i and α_i become

$$w_i = -\frac{\Gamma_{\max}}{2b} \quad (8.24)$$

$$\alpha_i = \frac{\Gamma_{\max}}{2bQ_{\infty}} \quad (8.24a)$$

and are constant along the wing span.

Another feature of the elliptic distribution is that the spanwise integral is simply half the area of an ellipse (with semi-axes $\Gamma_{\max}$ and $b/2$):

$$\int_{-b/2}^{b/2} \Gamma(y) dy = \frac{\pi}{2} \Gamma_{\max} \frac{b}{2} = \frac{\pi b}{4} \Gamma_{\max} \quad (8.25)$$

Consequently, the lift and the drag of the wing can be evaluated as

$$L = \rho Q_{\infty} \int_{-b/2}^{b/2} \Gamma(y) dy = \frac{\pi b}{4} \rho Q_{\infty} \Gamma_{\max} \quad (8.26)$$

$$D_i = \rho Q_{\infty} \int_{-b/2}^{b/2} \alpha_i \Gamma(y) dy = \alpha_i L = \left(\frac{\Gamma_{\max}}{2bQ_{\infty}} \right) \frac{\pi b}{4} \rho Q_{\infty} \Gamma_{\max} = \frac{\pi}{8} \rho \Gamma_{\max}^2 \quad (8.27)$$

The lift and drag coefficients become

$$C_L \equiv \frac{L}{(1/2)\rho Q_\infty^2 S} = \frac{\pi b}{2 S} \frac{\Gamma_{\max}}{Q_\infty} \quad (8.28)$$

$$C_{D_i} \equiv \frac{D_i}{(1/2)\rho Q_\infty^2 S} = \frac{\pi}{4 S} \frac{\Gamma_{\max}^2}{Q_\infty^2} = \frac{1}{\pi} \frac{S}{b^2} C_L^2 \quad (8.29)$$

Substituting the spanwise downwash (Eq. (8.24)) and the elliptic circulation distribution (Eq. (8.21)) into Eq. (8.16) yields

$$\frac{-2\Gamma_{\max}}{m_0(y)c(y)Q_\infty} \left[1 - \left(\frac{y}{b/2} \right)^2 \right]^{1/2} - \frac{\Gamma_{\max}}{2bQ_\infty} + \alpha(y) - \alpha_{L0}(y) = 0 \quad (8.30)$$

This equation provides the relation between the local chord $c(y)$ and the local angle of attack $\alpha(y)$ for the wing with the elliptic circulation distribution. If the chord $c(y)$ has an elliptic form, too, the constant $\Gamma_{\max}$ can be easily evaluated. Thus, assume

$$c(y) = c_0 \left[1 - \left(\frac{y}{b/2} \right)^2 \right]^{1/2} \quad (8.31)$$

where c_0 is the root chord. Substituting Eq. (8.31) into Eq. (8.30) cancels the elliptic variation and we have

$$\frac{-2\Gamma_{\max}}{m_0(y)c_0Q_\infty} - \frac{\Gamma_{\max}}{2bQ_\infty} + \alpha(y) - \alpha_{L0}(y) = 0 \quad (8.32)$$

For an elliptic planform with constant airfoil shape, all terms but $\alpha(y)$ in this equation are constant, and therefore this wing with an elliptic planform and load distribution is untwisted ($\alpha(y) = \alpha = \text{const.}$). The value of $\Gamma_{\max}$ is then

$$\Gamma_{\max} = \frac{2bQ_\infty(\alpha - \alpha_{L0})}{1 + 4b/m_0c_0} \quad (8.33)$$

The area S of the elliptic wing is

$$S = \pi \frac{c_0}{2} \frac{b}{2} \quad (8.34)$$

Also, it is common to define the wing aspect ratio $\mathcal{R}$ as

$$\mathcal{R} \equiv \frac{b^2}{S} \quad (8.35)$$

Using the $\mathcal{R}$ and the area S for the elliptic wing and substituting into Eq. (8.33), we obtain

$$\Gamma_{\max} = \frac{2bQ_\infty(\alpha - \alpha_{L0})}{1 + \pi \mathcal{R}/m_0} \quad (8.33a)$$

With this expression for $\Gamma_{\max}$ and using $m_0 = 2\pi$, the lift coefficient (Eq. (8.28)) becomes

$$C_L = \frac{2\pi}{1 + 2/\mathcal{R}} (\alpha - \alpha_{L0}) \equiv C_{L_\alpha} (\alpha - \alpha_{L0}) \quad (8.36)$$

Here C_{L_α} is the three-dimensional wing lift slope and the most important conclusion of this analysis is that this slope becomes less as the wingspan becomes smaller due to the induced downwash. This is illustrated by Fig. 8.7, where for a wing with given α_{L0} the effective

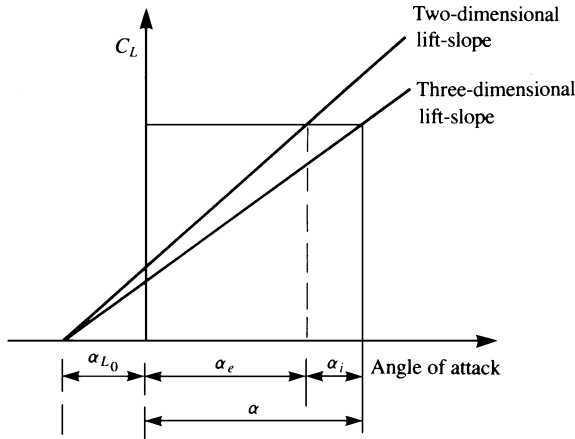


Figure 8.7 The reduction of lift slope for three-dimensional wings.

angle of attack is reduced by α_i according to Eq. (8.15). Consequently, for finite span wings, more incidence is needed to achieve the same lift coefficient as the wingspan decreases.

The induced drag coefficient is obtained by substituting Eq. (8.35) into Eq. (8.29):

$$C_{Di} = \frac{1}{\pi \mathcal{R}} C_L^2 \quad (8.37)$$

which indicates that as the wing aspect ratio increases the induced drag becomes smaller. Also, the induced drag for the finite elliptic wing will increase with a rate of C_L^2 as shown in Fig. 8.8.

The lift slope C_{L_α} versus $\mathcal{R}$ for the elliptic wing (Eq. (8.36)) is shown in Fig. 8.9. The lift slope of a two-dimensional wing is the largest (2π) and as the wingspan becomes smaller C_{L_α} decreases too.

The spanwise loading $L'(y)$ (lift per unit span) of the elliptic wing is obtained by using the Kutta–Joukowski theorem:

$$L'(y) = \rho Q_\infty \Gamma(y) = \rho Q_\infty \Gamma_{\max} \left[1 - \left(\frac{y}{b/2} \right)^2 \right]^{1/2} \quad (8.38)$$

Figure 8.10 shows an elliptic planform and the spanwise lift distribution $L'(y)$ that is elliptic too. As Eq. (8.24) indicated, the downwash of such a wing is constant, and, combined with

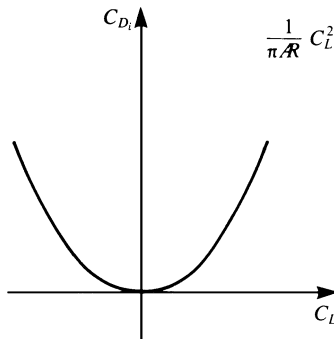


Figure 8.8 Lift polar for an elliptic wing.

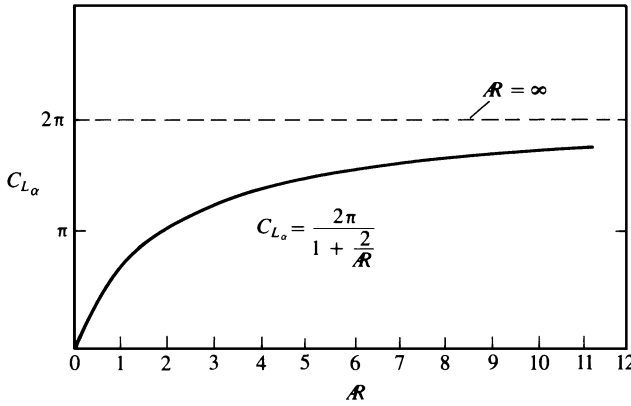


Figure 8.9 Variation of lift coefficient slope versus aspect ratio for thin elliptic wings.

the velocity induced by the bound vortex w_b , must be equal to the upwash of the free-stream $Q_\infty \alpha$ so that the combined normal velocity component is zero, according to Eq. (8.7). (Note that it is possible to have elliptic loading with other than an elliptic planform, but in that case, local twist or camber needs to be adjusted so that w_i will remain constant.)

The section lift coefficient C_l is defined by using the local chord from Eq. (8.31) and is

$$C_l \equiv \frac{L'(y)}{(1/2)\rho Q_\infty^2 c(y)} = \frac{2\Gamma_{\max}}{c_0 Q_\infty} = C_L$$

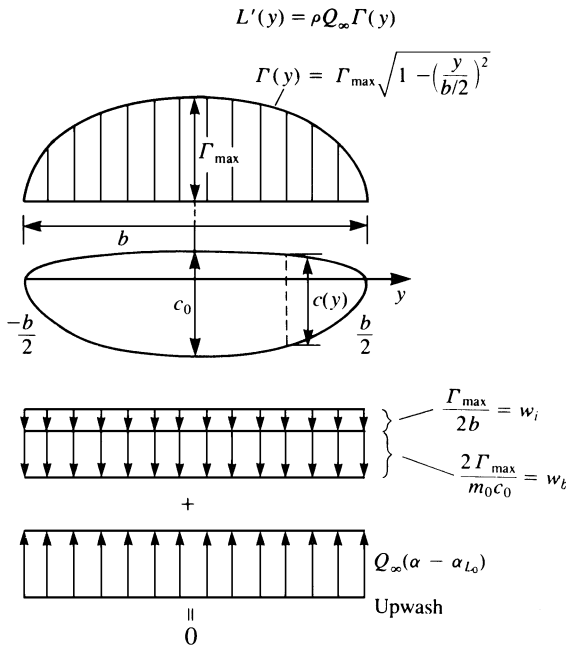


Figure 8.10 Chord and load distribution for a thin elliptic wing. Note that the induced downwash is constant and combined with the downwash of the bound vortex is equal to the free-stream upwash, resulting in zero velocity normal to the wing surface (Eq. (8.7)).

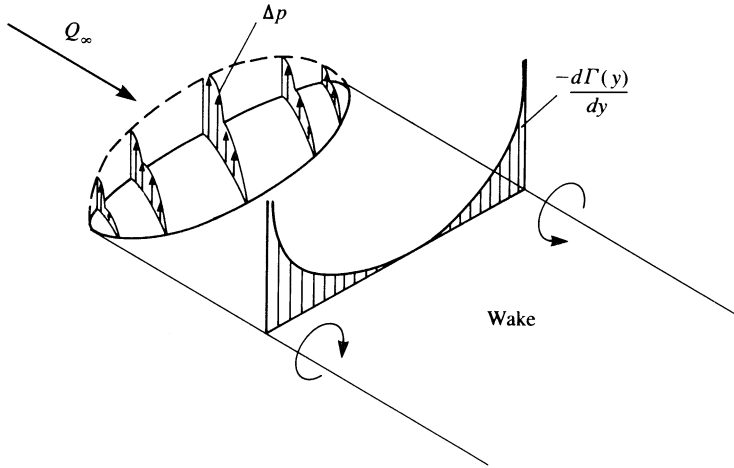


Figure 8.11 Schematic description of the pressure difference and wake vortex distribution of a thin elliptic wing.

Thus, for the elliptic wing, both section lift coefficient and wing lift coefficient are the same, that is,

$$C_l = \frac{2\pi}{1 + 2/R}(\alpha - \alpha_{L0}) \equiv C_{l_a}(\alpha - \alpha_{L0}) \quad (8.39)$$

Similarly, the section induced drag coefficient is

$$C_{d_i} \equiv \frac{L'(y)\alpha_i}{(1/2)\rho Q_\infty^2 c(y)} = \frac{1}{\pi} \frac{S}{b^2} C_l^2 = C_{D_i} \quad (8.40)$$

The strength of the circulation in the wake is simply the spanwise derivative of $\Gamma(y)$ (see Eq. (8.21)),

$$\frac{d\Gamma(y)}{dy} = -\frac{4\Gamma_{\max}}{b^2} \frac{y}{\sqrt{[1 - (y/(b/2))^2]}} \quad (8.41)$$

This spanwise wake vortex strength is shown schematically in Fig. 8.11. It is clear from this figure that near the wingtips, where $|\Gamma(y)/dy|$ is the largest, the wake vortex will be the strongest. Therefore, owing to the induced velocity at the wake it will roll up, mostly near the wingtips, to form two concentrated trailing vortices as shown by the flow visualization in Fig. 8.12. The induced effect of this wake rollup on $\Gamma(y)$ is assumed to be negligible in this model; but this effect can be investigated by the numerical methods of later chapters.

8.1.5 General Spanwise Circulation Distribution

A more general solution for the spanwise circulation $\Gamma(y)$ in Eq. (8.16) can be obtained by describing the unknown distribution in terms of a trigonometric expansion. Using the spanwise coordinate θ , as defined in Eq. (8.23), we select the following Fourier expansion:

$$\Gamma(\theta) = 2bQ_\infty \sum_{n=1}^{\infty} A_n \sin n\theta \quad (8.42)$$

The shapes of the first three symmetric terms in this expansion are shown schematically in



Figure 8.12 Flow visualization of the rollup of the trailing vortices behind an airplane (wing tip vortices made visible by ejecting smoke at the wing tips of a Boeing 727 airplane). (Courtesy of NASA.)

Fig. 8.13, and all terms fulfill Eq. (8.17) at the wingtips:

$$\Gamma(0) = \Gamma(\pi) = 0 \quad (8.43)$$

Substituting $\Gamma(\theta)$ and $d\Gamma(\theta)/dy$ into Eq. (8.16) yields

$$\begin{aligned} & \frac{-4b}{m_0(\theta)c(\theta)} \sum_{n=1}^{\infty} A_n \sin n\theta \\ & + \frac{-b}{2\pi} \int_{\pi}^0 \frac{\sum_{n=1}^{\infty} A_n n \cos n\theta_0 [1/(-b/2) \sin \theta_0][(-b/2) \sin \theta_0 d\theta_0]}{(b/2)(\cos \theta - \cos \theta_0)} \\ & + \alpha(\theta) - \alpha_{L0}(\theta) = \frac{-4b}{m_0(\theta)c(\theta)} \sum_{n=1}^{\infty} A_n \sin n\theta - \frac{1}{\pi} \int_0^{\pi} \frac{\sum_{n=1}^{\infty} n A_n \cos n\theta_0 d\theta_0}{\cos \theta_0 - \cos \theta} \\ & + \alpha(\theta) - \alpha_{L0}(\theta) = 0 \end{aligned} \quad (8.44)$$

Using Glauert's integral (Eq. (5.22)) for the second term gives us

$$\frac{-4b}{m_0(\theta)c(\theta)} \sum_{n=1}^{\infty} A_n \sin n\theta - \sum_{n=1}^{\infty} n A_n \frac{\sin n\theta}{\sin \theta} + \alpha(\theta) - \alpha_{L0}(\theta) = 0 \quad (8.44a)$$

Comparing this result with Eq. (8.16) indicates that the first term is $-\alpha_e$ and the second term is $-\alpha_i$:

$$\alpha_i(\theta) = \sum_{n=1}^{\infty} n A_n \frac{\sin n\theta}{\sin \theta} \quad (8.45)$$

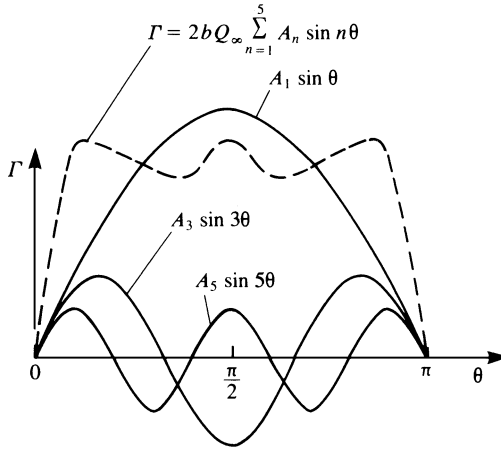


Figure 8.13 Sine series representation of symmetric spanwise circulation distribution $\Gamma(\theta)$, $n = 1, 3, 5$.

Therefore, the section lift and drag coefficients can be readily obtained:

$$C_l = \frac{\rho Q_\infty \Gamma(\theta)}{(1/2)\rho Q_\infty^2 c(\theta)} = \frac{4b}{c(\theta)} \sum_{n=1}^{\infty} A_n \sin n\theta \quad (8.46)$$

$$C_{d_i} = C_{l\alpha_i} = \frac{4b}{c(\theta)} \sum_{n=1}^{\infty} A_n \sin n\theta \left(\sum_{k=1}^{\infty} k A_k \frac{\sin k\theta}{\sin \theta} \right) \quad (8.47)$$

The wing aerodynamic coefficients are obtained by the spanwise integration of these section coefficients:

$$C_L = \int_{-b/2}^{b/2} \frac{C_l(y)c(y) dy}{S} = \frac{4b}{S} \int_0^\pi \sum_{n=1}^{\infty} A_n \sin n\theta \frac{b}{2} \sin \theta d\theta \quad (8.48)$$

$$C_{D_i} = \int_{-b/2}^{b/2} \frac{C_{d_i}(y)c(y) dy}{S} = \frac{2b^2}{S} \int_0^\pi \sum_{n=1}^{\infty} \sum_{k=1}^{\infty} k A_k A_n \sin k\theta \sin n\theta d\theta \quad (8.49)$$

Recall that

$$\int_0^\pi \sin n\theta \sin k\theta d\theta = \begin{cases} 0 & \text{for } n \neq k \\ \pi/2 & \text{for } n = k \end{cases} \quad (8.50)$$

and for the lift integral only the first term will appear. The lift coefficient becomes

$$C_L = \frac{\pi b^2 A_1}{S} = \pi \mathcal{R} A_1 \quad (8.51)$$

For the drag, only the terms where $n = k$ will be left:

$$C_{D_i} = \frac{\pi b^2}{S} \sum_{n=1}^{\infty} n A_n^2 = \pi \mathcal{R} \sum_{n=1}^{\infty} n A_n^2 \quad (8.52)$$

Using the results for the lift, we can rewrite this as

$$C_{D_i} = \frac{\pi^2 \mathcal{R}^2 A_1^2}{\pi \mathcal{R}} \left[1 + \sum_{n=2}^{\infty} \frac{n A_n^2}{A_1^2} \right] = \frac{C_L^2}{\pi \mathcal{R}} \left[1 + \sum_{n=2}^{\infty} \frac{n A_n^2}{A_1^2} \right] = (C_{D_i})_{\text{elliptic}} (1 + \delta_1) \quad (8.53)$$

where δ_1 includes the higher order terms for $n = 2, 3, \dots$ (only the odd terms are considered for a symmetric load distribution). This clearly indicates that for a given wing aspect ratio, the elliptic wing will have the lowest drag coefficient since for a generic wing planform $\delta_1 \geq 0$ and for the elliptic wing $\delta_1 = 0$.

Similarly, the lift coefficient for the general spanwise loading can be formulated as

$$C_L = \pi \mathcal{R} A_1 = m(\alpha - \alpha_{L_0}) \quad (8.54)$$

Assume that the wing is untwisted and therefore $\alpha - \alpha_{L_0} = \text{const}$. Following Glauert (Ref. 5.2, p. 142) we define an equivalent two-dimensional wing that has the same lift coefficient C_L . This wing is now set at an angle of attack $\alpha^* - \alpha_{L_0}$ such that

$$C_L = 2\pi(\alpha^* - \alpha_{L_0}) \quad (8.55)$$

The difference between these two cases is due to the wake-induced angle of attack, which is obtained from these two equations:

$$(\alpha - \alpha_{L_0}) - (\alpha^* - \alpha_{L_0}) = C_L \left[\frac{1}{m} - \frac{1}{2\pi} \right] \equiv \frac{C_L}{\pi \mathcal{R}} (1 + \delta_2) \quad (8.56)$$

and

$$1 + \delta_2 = \pi \mathcal{R} \left[\frac{\alpha - \alpha_{L_0}}{\pi \mathcal{R} A_1} - \frac{1}{2\pi} \right]$$

where $\delta_2 > 0$. Taking A_1 from this relation and substituting into Eq. (8.51) results in

$$C_L = \frac{2\pi(\alpha - \alpha_{L_0})}{1 + (2/\mathcal{R})(1 + \delta_2)} \quad (8.57)$$

Thus, for the elliptic wing $\delta_2 = 0$ and also its lift coefficient is higher than for wings with other spanwise load distributions.

8.1.6 Twisted Elliptic Wing

The spanwise loading of wings can be varied by introducing twist to the wing planform. To illustrate the effects of twist, consider a wing with an elliptic chord distribution. For this purpose let us rearrange Eq. (8.44a) such that

$$\sum_{n=1}^{\infty} A_n \sin n\theta \left[\frac{4b}{m_0(\theta)c(\theta)} + \frac{n}{\sin \theta} \right] = \alpha(\theta) - \alpha_{L_0}(\theta) \quad (8.58)$$

This is the governing equation for the coefficients for the circulation distribution for the general case that is described using lifting-line theory. Section 8.1.4 presents an exact solution for an untwisted elliptic planform wing (elliptic loading), but solutions for other cases must be obtained numerically using techniques that will be described in later sections. It is of interest to study the effect of wing twist on the solution for a particular geometry (geometric twist occurs for a spanwise variation of angle of attack and aerodynamic twist occurs for a spanwise variation of the zero-lift angle).

Filotas^{8.1} has found a closed-form solution for a wing with an elliptic planform and arbitrary twist and that solution will be presented in what follows. Consider an elliptic chord distribution as given in Eq. (8.31):

$$c = c_0 \sin \theta \quad (8.59)$$

and for simplicity let $m_0 = 2\pi$. Then Eq. (8.58) becomes

$$\sum_{n=1}^{\infty} A_n \sin n\theta \left[\frac{\mathcal{R}}{2} + n \right] = [\alpha(\theta) - \alpha_{L0}(\theta)] \sin \theta$$

where the aspect ratio of the elliptic wing is $\mathcal{R} = 4b/\pi c_0$. Note that the above equation is a Fourier series representation for the right-hand side whose coefficients are given by

$$A_n = \frac{2}{\pi} \frac{1}{\mathcal{R}/2 + n} \int_0^\pi [\alpha(\theta) - \alpha_{L0}(\theta)] \sin \theta \sin n\theta d\theta \quad (8.60)$$

To find the wing lift coefficient, the coefficient A_1 is obtained as

$$\begin{aligned} A_1 &= \frac{2}{\pi} \frac{1}{\mathcal{R}/2 + 1} \int_0^\pi [\alpha(\theta) - \alpha_{L0}(\theta)] \sin^2 \theta d\theta \\ &= \frac{4}{\pi} \frac{1}{\mathcal{R} + 2} \int_0^\pi [\alpha(\theta) - \alpha_{L0}(\theta)] \sin^2 \theta d\theta \end{aligned}$$

and the lift is obtained by using Eq. (8.51):

$$C_L = \frac{4\mathcal{R}}{\mathcal{R} + 2} \int_0^\pi [\alpha(\theta) - \alpha_{L0}(\theta)] \sin^2 \theta d\theta \quad (8.61)$$

Example:

As an example consider a wing with a linear twist where

$$\alpha(y) = \alpha \pm \alpha_0 \left| \frac{y}{b/2} \right| = \alpha \pm \alpha_0 |\cos \theta|$$

The effect of the twist can be analyzed by taking the variable part of $\alpha(y)$ only, and adding the contribution of the constant angle of attack later. Therefore, let

$$\alpha(y) = \alpha_0 |\cos \theta|$$

and use Eq. (8.60) to compute the coefficients A_n as

$$\begin{aligned} \frac{A_n}{\alpha_0} &= \frac{4}{\pi} \frac{1}{\mathcal{R}/2 + n} \int_0^{\pi/2} \cos \theta \sin \theta \sin n\theta d\theta \\ &= \frac{2}{\pi} \frac{1}{\mathcal{R}/2 + n} \int_0^{\pi/2} \sin 2\theta \sin n\theta d\theta \\ &= \frac{2}{\pi} \frac{1}{\mathcal{R}/2 + n} \left[\frac{\sin(n-2)\theta}{2(n-2)} - \frac{\sin(n+2)\theta}{2(n+2)} \right] \Big|_0^{\pi/2} \\ &= \frac{1}{\pi} \frac{1}{\mathcal{R}/2 + n} \left[\frac{\sin(n-2)\pi/2}{(n-2)} - \frac{\sin(n+2)\pi/2}{(n+2)} \right] \end{aligned}$$

Evaluating the individual coefficients for a wing with $\mathcal{R} = 6$ and for a twist of $\alpha = \alpha_0 |\cos \theta|$ and substituting into Eq. (8.42) yields

$$\Gamma(\theta) = \frac{2bQ_\infty\alpha_0}{\pi} \left[\frac{1}{3} \sin \theta + \frac{1}{5} \sin 3\theta - \frac{1}{42} \sin 5\theta + \frac{2}{225} \sin 7\theta - \dots \right]$$

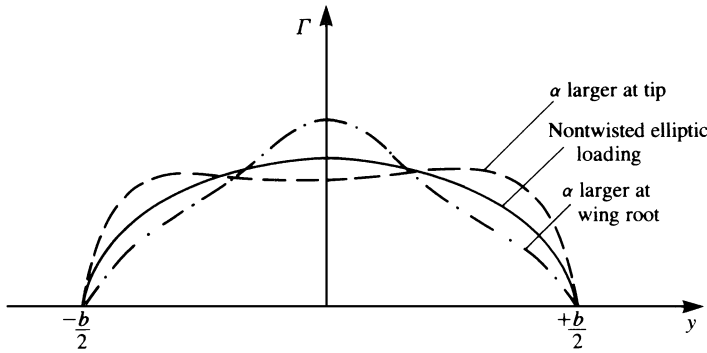


Figure 8.14 Effect of wing twist on the spanwise loading of an elliptic wing.

For a twist of $\alpha = \alpha_0(-|\cos \theta|)$, the circulation is

$$\Gamma(\theta) = \frac{2bQ_\infty\alpha_0}{\pi} \left[-\frac{1}{3} \sin \theta - \frac{1}{5} \sin 3\theta + \frac{1}{42} \sin 5\theta - \frac{2}{225} \sin 7\theta - \dots \right]$$

These results, combined with an additional constant angle of attack α are plotted schematically in Fig. 8.14, which shows that having a larger angle of attack at the tip will increase the load there. Similarly, larger angles of attack near the wing root will increase the loading there.

8.1.7 Conclusions from Lifting-Line Theory

The most important result of the lifting-line theory is its ability to establish the effect of wing aspect ratio on the lift slope and induced drag. Some of the more important conclusions are:

1. The wing lift slope $dC_L/d\alpha$ decreases as wing aspect ratio becomes smaller (as shown by Eq. (8.36) for an elliptic wing and by Eq. (8.57) for a wing with general spanwise circulation).
2. The induced drag of a wing increases as wing aspect ratio decreases (as shown by Eq. (8.37) for an elliptic wing and by Eq. (8.53) for a wing with general spanwise circulation).
3. A wing with elliptic loading will have the lowest induced drag and the highest lift, as indicated by Eqs. (8.53) and (8.57).
4. This theory also provides valuable information about the wing's spanwise loading and about the existence of the trailing vortex wake.
5. The theory is limited to small disturbances and large aspect ratio and, for example, Eq. (8.6), which requires that the wake be aligned with the local velocity, was not addressed at all (because of the small angle of attack assumption).
6. There are possible modifications to this theory, such as the addition of wing sweep (e.g., Weissinger^{8,2}). However, the study of wings with more complex geometry is difficult with this model whereas some of the more refined methods (introduced in the following chapters) are clearly more capable in dealing with this problem.
7. Using the results of this theory we must remember that the drag of a wing includes the induced drag portion (predicted by this model) plus the viscous drag, which must be taken into account.